SHRINCS specification

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Abstract

This document specifies **SHRINCS** (Stateful Hash-based Robust Integrated NIST-Compatible Signatures), a post-quantum signature scheme optimized for Bitcoin. SHRINCS combines efficient stateful hash-based signatures (unbalanced XMSS with WOTS+C) for normal operation with a stateless fallback (SPHINCS+C variant) for scenarios when state management failed. The scheme achieves minimal 324-byte signatures during normal stateful operation while maintaining the robustness of stateless signatures as a backup when state is corrupted or lost.

1 Motivation

Current Bitcoin signature schemes (ECDSA and Schnorr over secp256k1) are vulnerable to quantum attacks via Shor's algorithm. Hash-based signature schemes offer post-quantum security relying only on the security of cryptographic hash functions, which are believed to resist quantum attacks apart from Grover's quadratic speedup.

The key challenges for deploying hash-based signatures in Bitcoin are:

1. Signature size: NIST-standardized SLH-DSA signatures range from 7KB to 50KB, significantly larger than current 64-byte Schnorr signatures.
2. Stateful vs. Stateless tradeoff: Stateful schemes (XMSS) offer small signatures but require careful state management. Stateless schemes (SPHINCS+) are robust but produce large signatures.
3. Backup compatibility: Bitcoin users expect to restore wallets from seed phrases. Pure stateful schemes break this model since state cannot be statically backed up.

SHRINCS addresses these challenges by:

- Using efficient stateful signatures ($\approx$ 324 bytes for $q = 1$) during normal operation.
- Falling back to stateless signatures ($\approx$ 3-4 KB) when state is lost or corrupted.
- Allowing static seed backup while maintaining security guarantees.

2 Notation

Throughout this specification:

`||` denotes byte string concatenation

`[a:b]` denotes a substring from byte index `a` to `(b-1)`

`[0:a]` denotes first `a` bytes (truncation)

`←$` uniform random sampling

`n` is a the security parameter in bits

`H` denotes a cryptographic hash function

`Tw` denotes a tweakable hash function

`PRF` denotes a pseudorandom function

3 Overview

SHRINKS is designed in the way it can operate in two modes: **stateful** and **stateless**. Stateful mode is represented by the instance of Unbalanced XMSS (UXMSS) with a root public key `PK.sf`. An optimized variant of SLH-DSA is used to support the stateless operating mode (which is represented by the public key `PK.sl`). The root public key is a hashed concatenation of (`PK.sf`, `PK.sf`), which is being converted to the Bitcoin address.

During normal operation with intact state, the user generates signatures according to efficient stateful path. Each leaf of the UXMSS is represented by WOTS+C one time signature instance. UXMSS is right-skewed, which makes the first stateful signature the cheapest from the verification and size perspective (324 bytes) and adds the constant size (16 bytes) and one additional hashing operation for each next stateful signature.

If the state is corrupted (or the limit for stateful operations is met), the signature can continue operating with stateless mode. Stateless mode utilizes an optimized SLH-DSA variant with WOTS+ signatures in regular XMSS trees on each layer in the hypertree and FORS+C for few-time signature scheme. The general construction of SHRINCS signature is presented on the Figure 1.

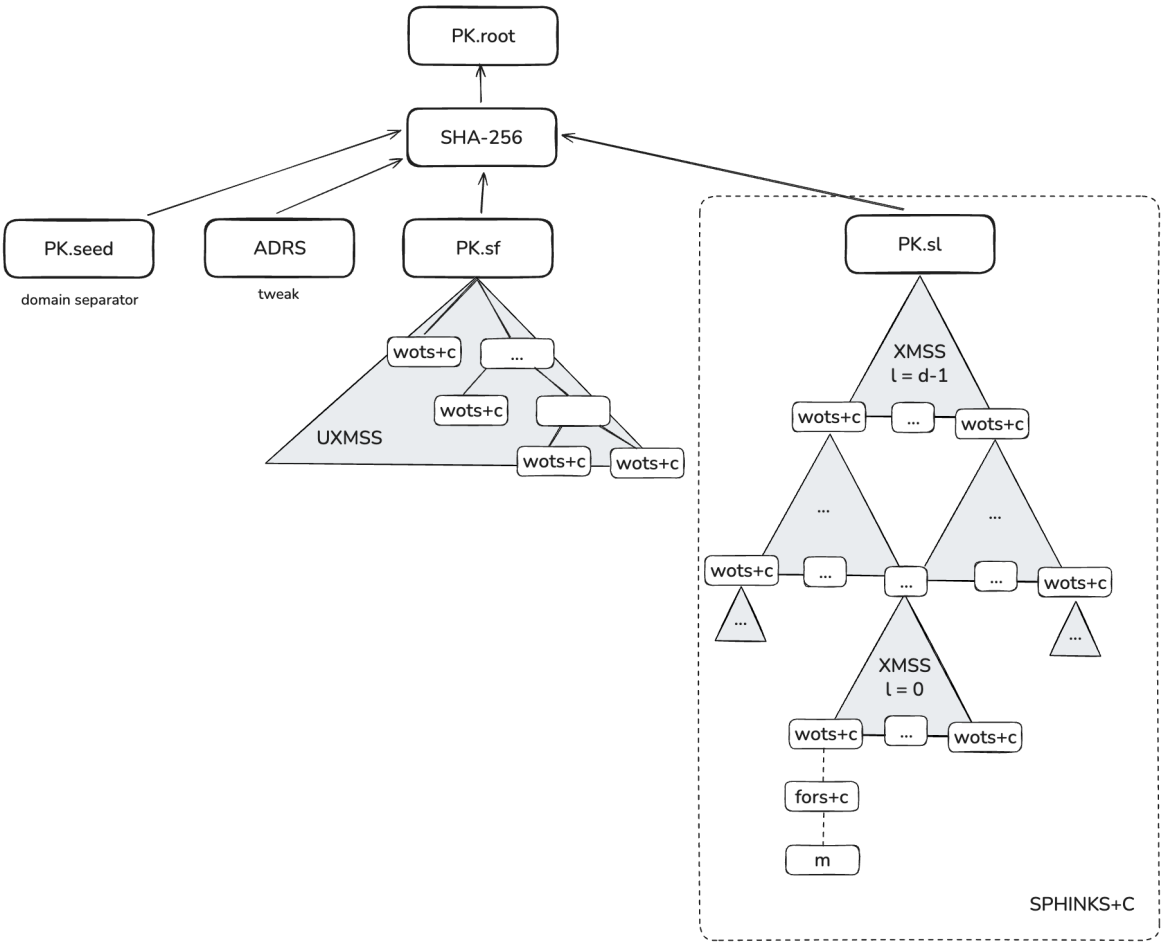

Figure 1: SHRINCS architecture

4 Parameters and Primitives

We define SHRINCS-SHA2-128+C as a SHRINCS instance preferable for Bitcoin. It operates with the following set of parameters:

Parameter	Value	Description
H	SHA-256	The underlying hash function
OTS	WOTS+C	The underlying one-time signature
FTS	FORS+C	The underlying few-time signature
n	16	Security parameter (bytes)
w	256	Winternitz parameter (8 bits per chain)
len	16	Message digest length, the number of WOTS+C chains
Swn	2040	Target sum for WOTS+C
hsf	15	Maximum stateful tree height
hsl	40	Maximum stateless hypertree height
d	5	Stateless hypertree layers
a	14	FORS+C tree height
k	11	Number of FORS+C trees
c	4	Counter size for WOTS+C grinding (bytes)
R	32	Randomness (bytes)

4.1 SHRINCS Key Material

SHRINCS keys are derived from a single random seed:

```
seed ←$ {0,1}^(3n)
```

Seed allows to derive a private key `SK ← (SK.seed, SK.prf, PK.seed, PK.sf, PK.sl, PK.root)`:

```
SK.seed ← seed[0:n]  // Seed for key derivation
SK.prf ← seed[n:2n]  // Secret needed for PRF_msg
PK.seed ← seed[2n:3n] // Domain separator for different SHRINCS key pairs
PK.sf ← compute_stateful_root(PK.seed, ADRS, SK.seed) // Public stateful key, see Section X
PK.sl ← compute_stateless_root(PK.seed, ADRS, SK.seed) // Public stateless key, see Section X
PK.root ← H(PK.seed, ADRS, PK_sf, PK_sl) // Root public key
```

where `ADRS` is a context address (see Section 4.1).

The size of the secret key is `6n` bytes. A public key consists of two values `PK ← (PK.seed, PK.root)` and has a size of `2n` bytes.

4.2 Hash Function

SHRINCS uses SHA-256 truncated to `n` bits for all internal hash operations.

```
H(m) = SHA-256(m)[0:n]
```

4.3 Tweakable Hash Function

To prevent multi-target attacks and ensure domain separation between the various components, we employ the tweakable hash function construction defined in FIPS 205:

```
Tw(PK.seed, ADRS, m) = H(PK.seed || ADRS || m)
```

This usage of tweaked hashing is critical. It ensures that a hash computed at one position in the Merkle tree cannot be substituted for a hash at another position, effectively mitigating generic tree-grafting attacks.

4.4 Pseudorandom Functions

We use PRF for deterministic key derivation:

```
PRF(PK.seed, ADRS, SK.seed) = Tw(PK.seed || ADRS || SK.seed)[0:n]
```

Additionally we use a pseudorandom function to generate the randomizer `R` for the randomized hashing of the message to be signed.

```
PRF_msg(SK.prf, PK.seed, opt_rand, m) = HMAC-SHA-256(SK.prf, PK.seed || opt_rand ||
                                        m)[0:n],
```

where `opt_rand` is an optional randomness (`n` bytes), can be `PK.seed` for deterministic signing.

4.5 Message Hash Function

To generate the digest of the message to be signed we use the following function:

```
H_msg(ADRS, R, PK.seed, PK.root, m) = H(ADRS || R || PK.seed || PK.root || m)
```

5 SHRINCS-SHA2-128s Specification

5.1 Tweak Structure and Domain Separation

Tweaks are structured as follows to ensure uniqueness across all scheme components:

```
ADRS = layer || tree_address || type || words
```

where:

`layer`: (4 bytes) layer in hypertree (`0` = bottom, `d-1` = max)

`tree_address`: (12 bytes) tree address within layer

`type`: (4 bytes) address type

`words`: (12 bytes) type-dependent 12 bytes (Appendix B)

5.1.1 Address Types

`0x00`: Stateful WOTS+C chains (`SF_WOTS_CHAIN`)

`0x01`: Stateful tree nodes (`SF_TREE`)

`0x02`: Stateless WOTS+C chains (`SL_WOTS_CHAIN`)

`0x03`: Stateless tree nodes (`SL_TREE`)

`0x04`: FORS (`SL_FORS`)

`0x05`: Message hash (`H_MSG`)

`0x06`: Pseudorandom function (`PRF`)

`0x07`: Root public key (`ROOT`)

5.1.2 Address Manipulation Functions

```
setLayerAddress(ADRS, layer):
    ADRS[0:4] ← toByte(layer, 4)

setTreeAddress(ADRS, tree_addr):
    ADRS[4:16] ← toByte(tree_addr, 12)

setTypeAndClear(ADRS, type):
    ADRS[16:20] ← toByte(type, 4)
    ADRS[20:32] ← toByte(0, 12)    // Clear words 5-7
```

```
setKeyPairAddress(ADRS, keypair):
    ADRS[20:24] ← toByte(keypair, 4)

setChainAddress(ADRS, chain):
    ADRS[24:28] ← toByte(chain, 4)

setHashAddress(ADRS, hash):
    ADRS[28:32] ← toByte(hash, 4)

setTreeHeight(ADRS, height):
    ADRS[24:28] ← toByte(height, 4)

setTreeIndex(ADRS, index):
    ADRS[28:32] ← toByte(index, 4)
```

Oleks: TODO: type-dependent words

5.2 WOTS+C. The Stateful Leaf

The standard Winternitz One-Time Signature (WOTS+) is the building block of most modern hash-based schemes. It works by encoding the message into a series of small integers and using them to iterate hash chains. A significant portion of a WOTS+ signature size (approx 20%) is dedicated to a "Checksum," which is necessary to prevent a forgery attack where an adversary extends the hash chains forward.

WOTS+C (Compressed WOTS+) eliminates the transmission of this checksum entirely, achieving a smaller signature size. In WOTS+, the checksum C is calculated from the message digest D.

$$C = \sum_{i=1}^{\ell} (w - 1 - d_i)$$

This checksum is then signed using additional chains. In WOTS+C, instead of calculating a checksum for a fixed message digest, the signer modifies the message digest until it possesses a specific, predetermined checksum property. This is done by appending a counter `count` and a random salt `R` to the message before hashing. The signer increments cnt until the resulting hash digest D satisfies a global target condition.

5.2.1 The Target Condition

For SHRINCS-SHA2-128s, we use the Winternitz parameter $w = 256$. This means each chain processes 8 bits (1 byte) of the digest. The message digest length is 128 bits (16 bytes), so $\ell = 16$. The values d_i are treated as integers in w.

We define the expected value of the checksum. If the hash output is uniform, the expected value of a single digit is $(w - 1)/2 = 127.5$. For 16 digits, the expected sum is $16 \times 127.5 = 2040$.

The condition for a valid WOTS+C message digest D is:

$$\sum_{i=0}^{15} (255 - d_i) = 2040$$

By enforcing this condition during signing (including grinding), the verifier can simply assume the checksum is 2040 without receiving it. If the reconstructed digest does not sum to 2040, the signature is invalid.

5.2.2 Parameters

Parameter	SHRINCS-SHA2-128s
n	128 bits
w	256
l	16 chains
Swn	2040
r	4 bytes

5.2.3 Base-w Representation

```
Algorithm: base_w(m, w, out_len)
Input:
    m: byte string
    w: Winternitz parameter (256)
    out_len: output length (16)
Output:
    b_w: array of out_len integers in [0, w-1]

1. in_idx ← 0
2. bits ← 0
3. total ← 0
4. b_w ← []
5. for i from 0 to out_len - 1:
    a. if bits == 0:
        total ← m[in_idx]
        in_idx ← in_idx + 1
        bits ← 8
    b. bits ← bits - log2(w)
    c. base_w[i] ← (total >> bits) AND (w - 1)
6. return b_w
// for w = 256, each byte becomes one base-w digit
```

5.2.4 Chain Function

```
Algorithm: chain(m, start, steps, PK.seed, ADRS)
Input:
    m: (n/8)-byte input
    start: starting index in chain
    steps: number of iterations
    PK.seed: public seed
    ADRS: tweak
Output:
    (n/8)-byte chain output

1. if steps == 0:
    return m
2. tmp ← chain(m, start, steps - 1, PK.seed, ADRS)
3. setHashAddress(ADRS, start + steps - 1)
4. return Tw(PK.seed, ADRS, tmp)
```

5.2.5 Key Generation

```
Algorithm: wots_PKgen(SK.seed, PK.seed, ADRS)
Input:
    SK.seed: secret seed
    PK.seed: public seed
```

```
    ADRS: tweak
Output:
    pk: compressed WOTS+C public key ((n/8) bytes)

1. setTypeAndClear(ADRS, WOTS_PRF)
2. setKeyPairAddress(ADRS, ADRS.getKeyPairAddress())
3. tmp ← []
4. for i from 0 to l - 1:
    a. setChainAddress(ADRS, i)
    b. sk_i ← PRF(SK.seed, ADRS)
    c. setTypeAndClear(ADRS, WOTS_HASH)
    d. setKeyPairAddress(ADRS, ADRS.getKeyPairAddress())
    e. setChainAddress(ADRS, i)
    f. tmp[i] ← chain(sk_i, 0, w - 1, PK.seed, ADRS)
5. setTypeAndClear(ADRS, WOTS_PK)
6. setKeyPairAddress(ADRS, ADRS.getKeyPairAddress())
7. pk ← Tw(PK.seed, ADRS, tmp[0] || ... || tmp[l-1])
8. return pk
```

5.2.6 Messsage Digest Grinding

```
Algorithm: wots_grind(m, R, PK.seed, PK.root)
Input:
    m: message to sign
    R: randomness
    counter: counter for grinding
    PK.seed, PK.root: public key components
Output:
    msg: array of l integers
    counter: final value used for grinding

1. counter ← 0
2. for counter from 0 to 2^32:
    a. digest ← H_msg(R || counter, PK.seed, PK.root, m)[0:n]
    b. msg ← base_w(digest, w, l)
    c. sum ← 0
    d. for i from 0 to l - 1:
        sum ← sum + msg[i]
    e. if (sum == Swn)
        return (msg, counter)
    f. counter ← counter + 1
    g. if counter == 2^32:
        abort with error // R must be resampled
```

Expected iterations value is ≈ 2000-3000 for SHRINCS-SHA2-128s with `Swn = 2040`.

5.2.7 Signature Generation

```
Algorithm: wots_sign(m, SK.seed, SK.prf, PK.seed, PK.root, ADRS)
Input:
    m: message
    SK.seed: secret seed
    SK.prf: PRF key for randomness
    PK.seed: public seed
    PK.root: public key root
    ADRS: tweak
Output:
```

```
    sig: WOTS+C signature instance

1. R ← PRF_msg(SK.prf, ADRS, m)
2. (msg, counter) ← wots_grind(m, R, PK.seed, PK.root)
3. sig ← R || counter
4. setTypeAndClear(ADRS, WOTS_PRF)
5. for i from 0 to l - 1:
    a. setChainAddress(ADRS, i)
    b. sk_i ← PRF(SK.seed, ADRS)
    c. setTypeAndClear(ADRS, WOTS_HASH)
    d. setChainAddress(ADRS, i)
    e. sig ← sig || chain(sk_i, 0, msg[i], PK.seed, ADRS)
6. return sig
```

5.2.8 Signature Verification (Public Key Recovery)

```
Algorithm: wots_pkFromSig(sig, m, PK.seed, PK.root, ADRS)
Input:
    sig: WOTS+C signature
    m: signed message
    PK.seed: public seed
    PK.root: public key root
    ADRS: tweak
Output:
    pk: recovered public key (n bytes), or $\bot$ if invalid

1. R ← sig[0:32]
2. counter ← sig[32:36]
3. (msg, valid) ← wots_digest(m, R, counter, PK.seed, PK.root) // like grinding but with known c
4. if not valid:
    return $\bot$
5. tmp ← []
6. setTypeAndClear(ADRS, WOTS_HASH)
7. for i from 0 to l - 1:
    a. setChainAddress(ADRS, i)
    b. sig_i ← sig[36 + i*n : 36 + (i+1)*n]
    c. tmp[i] ← chain(sig_i, msg[i], w - 1 - msg[i], PK.seed, ADRS)
8. setTypeAndClear(ADRS, WOTS_PK)
9. pk ← Tw(PK.seed, ADRS, tmp[0] || ... || tmp[l-1])
10. return pk
```

5.3 UXMSS. The Stateful Tree

Standard XMSS uses a balanced binary tree, which minimizes the maximum path length but imposes a logarithmic cost on every signature. For `N` signatures, every signature has size proportional to `log_2(N)`. SHRINCS utilizes an Unbalanced Merkle Tree (specifically a "Right-Skewed" tree) to optimize for the typical Bitcoin use case: a wallet that performs very few transactions in its lifecycle (we are targeting `N<=10`).

In this topology the `q`-th signature requires a path of length `q+1`. This linear growth is superior to logarithmic growth for small `q`. For `q=0`, the path is just 16 bytes. Even for `q=10`, the path is $11 \times 16 = 176$ bytes, which is extremely small compared to the fixed ≈ 2.5KB of a balanced tree or the ≈ 8KB of SLH-DSA.

The rightmost leaf at maximum depth contains the hash of the stateless public key, enabling the fallback mechanism.

5.3.1 UXMSS Authentication Path

For the q-th signature (q = count after signing), the authentication path contains q nodes:

```
auth_path = [node_0, node_1, ..., node_q-1]
```

Root computation from auth path:

```
Algorithm: root_from_auth(pk, auth, q, PK.seed, ADRS)
Input:
    pk: OTS public key at leaf position (q-1)
    auth: authentication path (q nodes)
    q: signature count after signing
    PK.seed: public seed
    ADRS: tweak
Output:
    root: computed tree root

1. node ← pk
2. setTypeAndClear(ADRS, TREE)
3. for i from 0 to q - 1:
    a. setTreeHeight(ADRS, i + 1)
    b. setTreeIndex(ADRS, 0)
    c. if i < q - 1:
        // auth[i] is left sibling
        node ← Tw(PK.seed, ADRS, auth[i] || node)
    d. else:
        // node is left child
        node ← Tw(PK.seed, ADRS, node || auth[i])
4. return node
```

5.4 SLH-DSA. The Stateless Fallback

5.4.1 Parameters

Parameter	SHRINCS-SHA2-128s
h	63 (total tree height)
d	7 (number of hypertree layers)
h'	9 (height per layer)
a	12 (FORS tree height)
k	14 (number of FORS trees)
w	256
n	128

5.5 Size

5.5.1 Stateful signature size

The WOTS+C component is 292 bytes: randomness (32 bytes) + counter (4 bytes) + chain values (256 bytes). The authentication path for signature q (count after signing) is `q * (n/8) = q * 16` bytes. The stateless public key `PK_sl` adds `(n/8) = 16` bytes.

`sig_sl = 292 + q * 16 + 16 = 308 + q * 16` bytes,

where q is the signature count after signing:

q	Auth path	Total
1	16 bytes	324 bytes
2	32 bytes	340 bytes
3	48 bytes	356 bytes
10	160 bytes	468 bytes

5.5.2 Stateless signature size

5.6 SHRINCS Core Algorithms

5.6.1 Key Generation

```
Algorithm: SHRINCS.KeyGen()
Output:
    SK: secret key
    PK: public key
    state: initial state

1. seed ←$ {0,1}^384
2. SK.seed ← seed[0:16]
3. SK.prf ← seed[16:32]
4. PK.seed ← seed[32:48]
5. ADRS ← new_ADRS()

// Generate stateful public key (UXMSS root)
6. setLayerAddress(ADRS, 0)
7. setTreeAddress(ADRS, 0)
8. PK_sf ← compute_initial_stateful_root(SK.seed, PK.seed, ADRS)

// Generate stateless public key (SPHINCS+C root)
9. PK_sl ← sphincs_root(SK.seed, PK.seed, ADRS)

// Combined params
10. PK.root ← H(0x08 || PK.seed || PK_sf || PK_sl)
11. SK ← (SK.seed, SK.prf, PK.seed, PK.root, PK_sf, PK_sl)
12. PK ← (PK.seed, PK.root)
13. state ← { q: 0, valid: true }
14. return (SK, PK, state)
```

5.6.2 Recovery from Seed

```
Algorithm: SHRINCS.Restore(seed)
Input:
    seed: master seed
Output:
    SK: secret key
    PK: public key
    state: state "LOST"

1. SK.seed ← seed[0:16]
2. SK.prf ← seed[16:32]
3. PK.seed ← seed[32:48]
4. ADRS ← new_ADRS()
5. PK_sf ← compute_max_stateful_root(SK.seed, PK.seed, ADRS)
6. PK_sl ← sphincs_root(SK.seed, PK.seed, ADRS)
7. PK.root ← H_n(0x08 || PK.seed || PK_sf || PK_sl)
8. SK ← (SK.seed, SK.prf, PK.seed, PK.root, PK_sf, PK_sl)
9. PK ← (PK.seed, PK.root)
10. state ← { q: $\bot$, valid: false }
11. return (SK, PK, state)
```

5.6.3 Statefull signing

5.6.4 Stateless signing

5.6.5 Statefull verification

5.6.6 Stateless verification

5.7 Data structures

5.7.1 Secret Key Format

5.7.2 Public Key Format

5.7.3 Stateful Signature Format

5.7.4 Stateless Signature Format